

Aditya Chunduri

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EDUCATION

University of Southern California

Master of Science in Applied Data Science

Coursework: Machine Learning, Information Retrieval, Data Mining, Analysis of Algorithms, Data Management, Research Methods

Los Angeles, CA

Jan 2024 - Dec 2025

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering

Chennai, India

Jun 2019 - Jul 2023

Specialization: Artificial Intelligence and Machine Learning

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, Bash

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face

MLOps & Cloud: Docker, Git, Flask, AWS, GitHub Actions, CI/CD

Data & Tools: PostgreSQL, ChromaDB, Neo4j, OpenCV, Pandas, NumPy

EXPERIENCE

University of Southern California

Research Assistant

Los Angeles, CA

Jun 2023 - Dec 2023

- Engineered CVD-Masks pipeline for artery-vein classification, optimizing U-Nets to achieve state-of-the-art Dice scores of 94%.
- Built automated RMHAS image scraper with quality controls, reducing manual data curation time by 40% for 10,000+ medical images.
- Developed modular PyTorch training pipelines to accelerate experimentation cycles by 30% and streamline fine-tuning.

SSN College of Engineering

Research Intern

Remote

Jun 2021 - Jul 2021

- Designed scalable computer vision pipelines for real-time object tracking, processing 30+ fps for high-quality video dataset generation.
- Fine-tuned YOLO-based CNN architectures to improve object detection robustness by 15% across 5 environmental conditions.
- Implemented 10+ reusable deep learning modules in TensorFlow and PyTorch to standardize model deployment workflows.

PROJECTS

Automated GenAI Media Pipeline

Akashic Tree (Personal Project, Role: Developer, Team of 1)

Python, GenAI

Aug 2025 - Present

- Engineered Quality Control Protocols, enforcing strict constraints to filter 10,000+ generated outputs, achieving a 99% safety compliance rate.
- Designed a Cost-Optimized Architecture with early-exit gates, preventing wasted compute on low-fidelity outputs and reducing inference costs by 45%.
- Orchestrated complex multimodal workflows, processing 500+ GB of unstructured data monthly while maintaining 99.9% pipeline uptime.

AI Health Journal

Mental Health Assistant (Personal Project, Role: Developer, Team of 1)

Python, Flask, Ollama, ChromaDB

Jan 2025 - Aug 2025

- Architected local-first RAG system using Ollama for privacy-preserving 7B parameter LLM inference on consumer hardware.
- Implemented Direct Preference Optimization (DPO) fine-tuning, improving response alignment and safety metrics by 25%.
- Optimized retrieval latency by 35% via ChromaDB indexing, enabling sub-second context injection for 10k+ documents.

Model-Behavior-Lab

LLM Benchmarking Platform (Personal Project, Role: Developer, Team of 1)

Python, Ollama, Plotly, CI/CD

Sep 2024 - Dec 2024

- Built automated CI/CD pipelines to evaluate 10+ local LLMs across 5,000+ reasoning and hallucination test cases.
- Developed interactive Plotly dashboards, reducing manual evaluation time by 50% and visualizing real-time trends.

MetaLearnML

AutoML System (Personal Project, Role: Developer, Team of 1)

Python, Neo4j, Scikit-learn

May 2024 - Aug 2024

- Engineered meta-learning AutoML engine to predict optimal architectures for 20+ datasets, reducing search time by 60%.
- Integrated Neo4j graph database to map 500+ dataset features to historical performance for smart recommendations.

PUBLICATIONS

Wind Power Analysis Using Machine Learning

Co-Author (Role: Researcher, Team of 4)

IJRASET

Aug 2023

- Proposed neural network framework integrating numerical weather data to predict wind energy, improving reliability across 5+ datasets.